SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Design and Analysis of Algorithms **COURSE CODE: CSA06**

COURSE OUTCOME COVERED

CO2: Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1,BL2,BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks: 50

Annexure A

Scenario Based Assessment

Code of Conduct: I C .pavani/(192524260)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:C.PAVANI

Annexure A

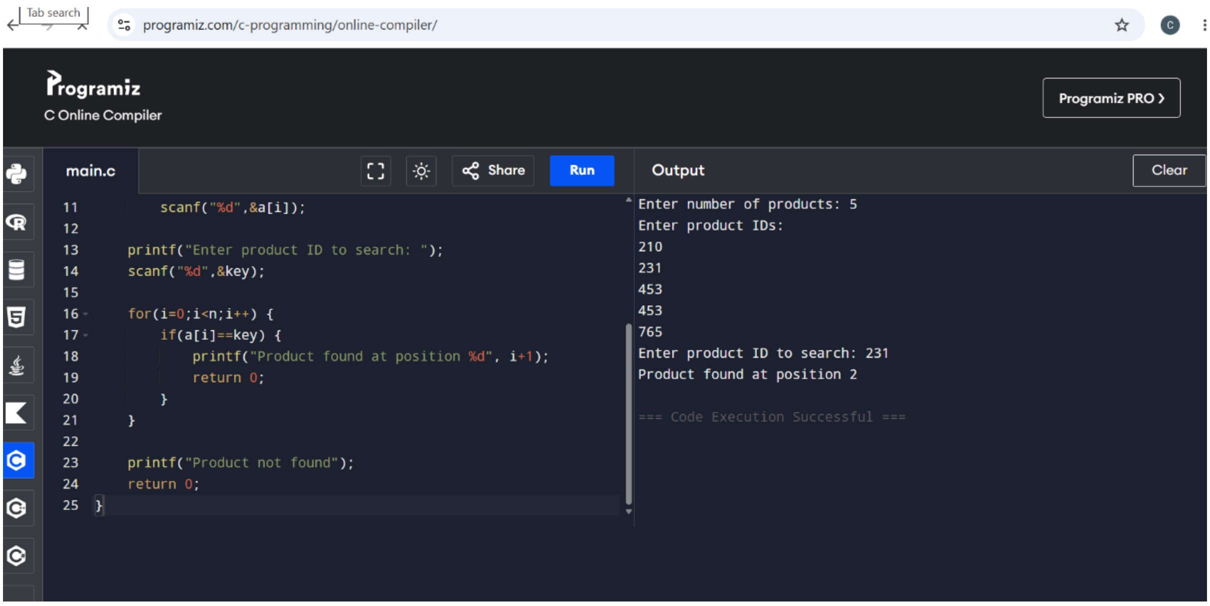
Scenario-Based Assessment

QUESTION 16:

CODE:

```
#include <stdio.h>
int main() {
    int a[100], n, key, i;
    printf("Enter number of products: ");
    scanf("%d", &n);
    printf("Enter product IDs:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    printf("Enter product ID to search: ");
    scanf("%d",&key);
    for(i=0;i<n;i++) {
        if(a[i]==key) {
            printf("Product found at position %d", i+1);
```

```
            return 0;
        }
    }
    printf("Product not found");
    return 0;
}
```

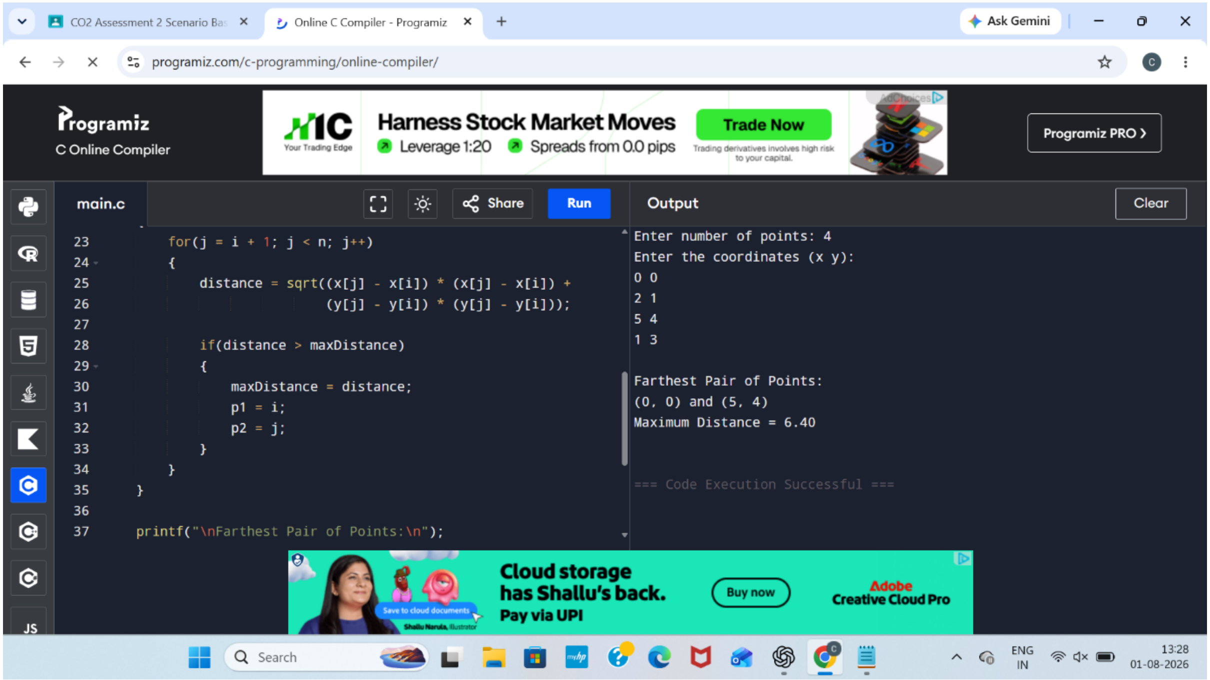


QUESTION 17:

CODE:

```
#include <stdio.h>
int main() {
    int x[10], y[10], n, i, j;
    float d, max = 0;
    printf("Enter number of points: ");
    scanf("%d",&n);
    for(i=0;i<n;i++) {
        scanf("%d%d",&x[i],&y[i]);
    }
    for(i=0;i<n;i++) {
        for(j=i+1;j<n;j++) {
            d = (x[i]-x[j])*(x[i]-x[j]) + (y[i]-y[j])*(y[i]-y[j]);
```

```
            if(d > max)
                max = d;
        }
    }
    printf("Maximum squared distance = %.2f", max);
    return 0;
}
```

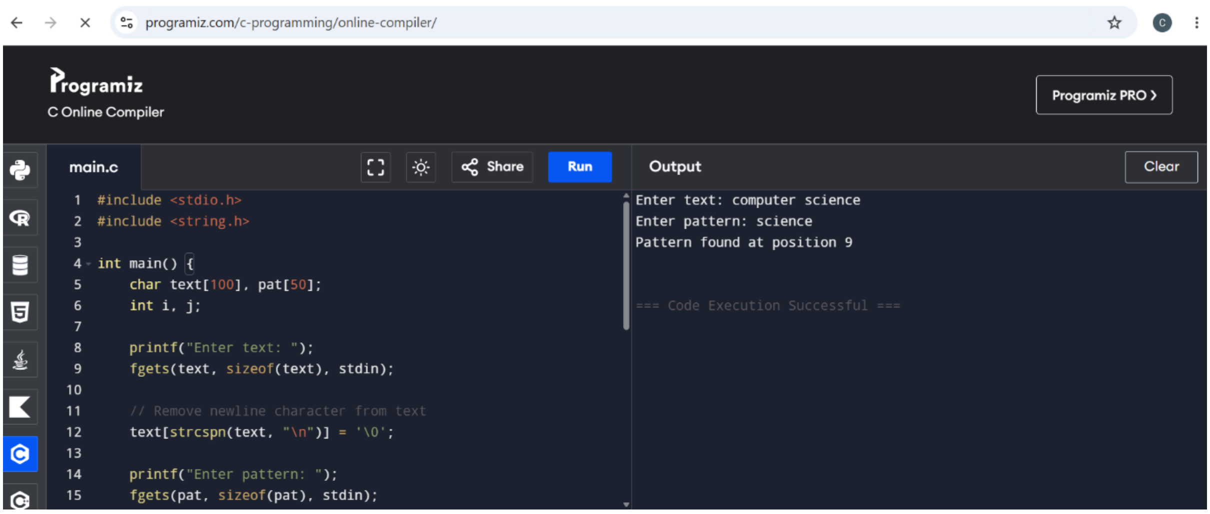


QUESTION 18:

CODE:

```
#include <stdio.h>
#include <string.h>
int main() {
    char text[100], pat[50];
    int i, j;
    printf("Enter text: ");
    gets(text);
    printf("Enter pattern: ");
    gets(pat);
```

```
  for(i=0;i<=strlen(text)-strlen(pat);i++) {
    for(j=0;j<strlen(pat);j++) {
      if(text[i+j]!=pat[j])
        break;
    }
    if(j==strlen(pat)) {
      printf("Pattern found at position %d", i);
      return 0;
    }
  }
  printf("Pattern not found");
  return 0;
}
```

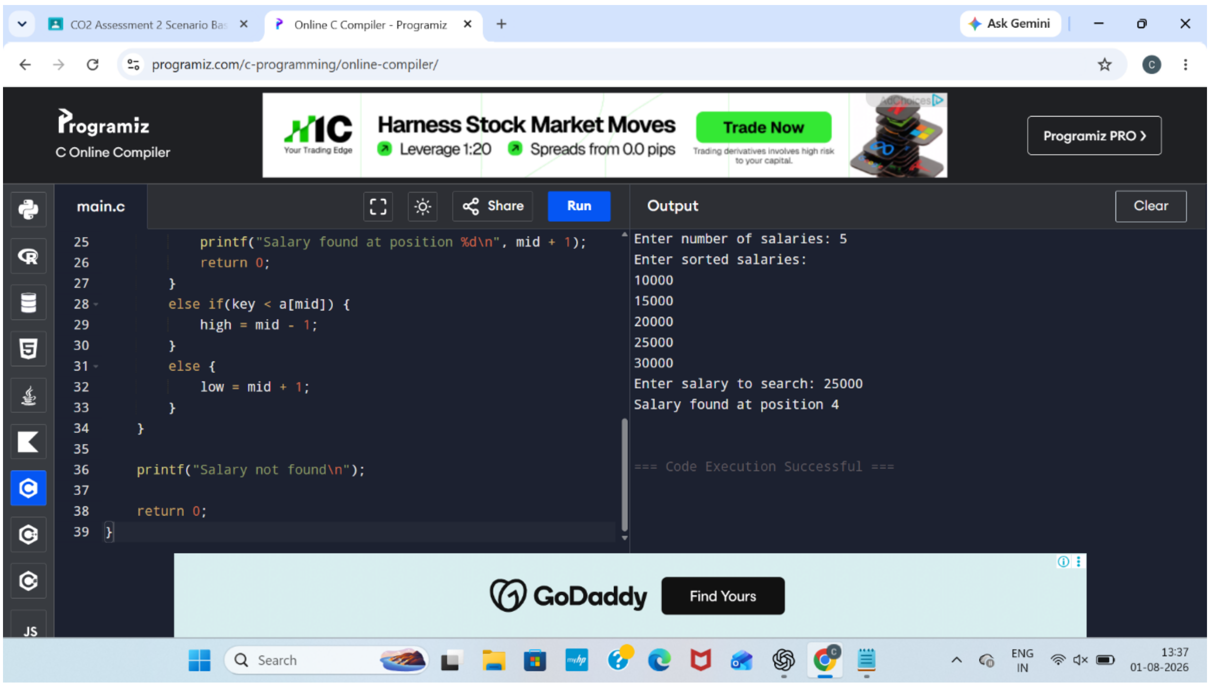


QUESTION 19:

CODE:

```
#include <stdio.h>
int main() {
  int a[100], n, key;
  int low=0, high, mid;
  printf("Enter number of salaries: ");
  scanf("%d",&n);
  printf("Enter sorted salaries:\n");
  for(int i=0;i<n;i++)
    scanf("%d",&a[i]);
```

```
    printf("Enter salary to search: ");
    scanf("%d",&key);
    high = n-1;
    while(low<=high) {
        mid = (low+high)/2;
        if(a[mid]==key) {
            printf("Salary found at position %d", mid+1);
            return 0;
        }
        else if(key<a[mid])
            high=mid-1;
        else
            low=mid+1;
    }
    printf("Salary not found");
    return 0;
}
```

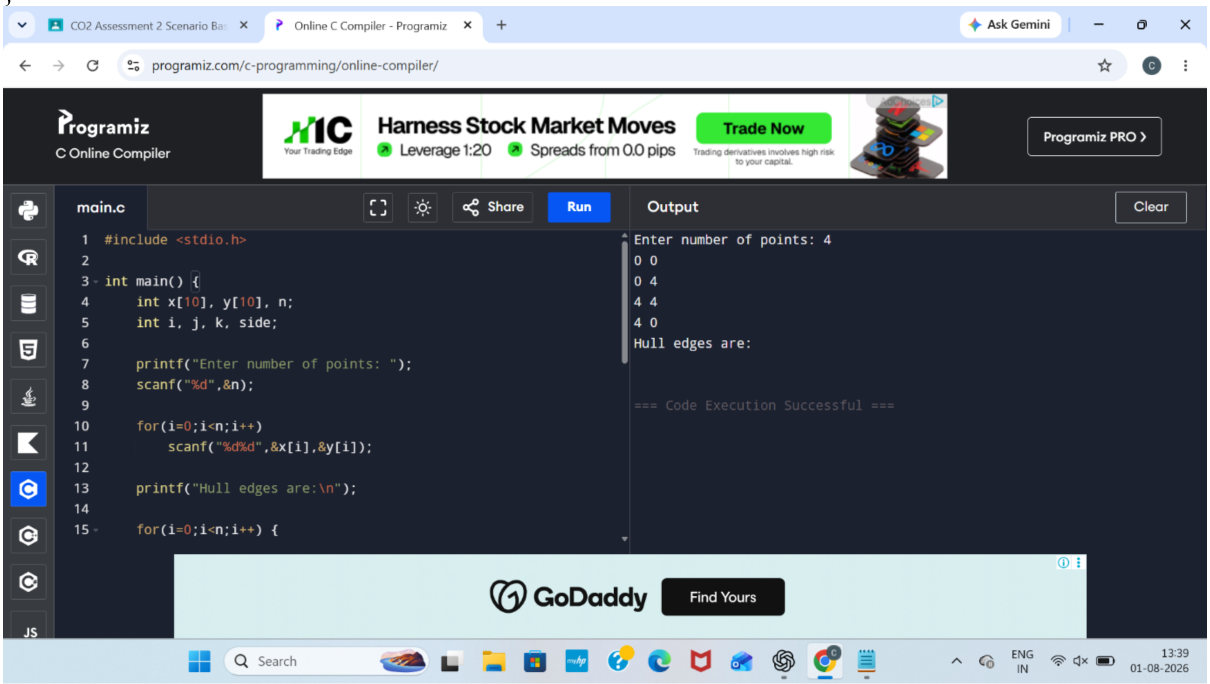


QUESTION 20:

CODE:

```
#include <stdio.h>
int main() {
    int x[10], y[10], n;
    int i, j, k, side;
    printf("Enter number of points: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d%d",&x[i],&y[i]);
    printf("Hull edges are:\n");
    for(i=0;i<n;i++) {
        for(j=i+1;j<n;j++) {
            side = 0;
            for(k=0;k<n;k++) {
                int val = (x[j]-x[i])*(y[k]-y[i]) - (y[j]-y[i])*(x[k]-x[i]);
                if(val>0)
                    side++;
                else if(val<0)
                    side--;
            }
            if(side==n || side==-n)
                printf("(%d,%d) -> (%d,%d)\n", x[i], y[i], x[j], y[j]);
        }
    }
    return 0;
```

}


CO2 Assessment 2 Scenario Ba
Online C Compiler - Programiz
Ask Gemini
programiz.com/c-programming/online-compiler/
Programiz
C Online Compiler
Harness Stock Market Moves
Leverage 1:20
Spreads from 0.0 pips
Trade Now
Programiz PRO
main.c
Share
Run
Output
Clear
#include <stdio.h>
int main() {
int x[10], y[10], n;
int i, j, k, side;
printf("Enter number of points: ");
scanf("%d",&n);
for(i=0;i<n;i++)
scanf("%d%d",&x[i],&y[i]);
printf("Hull edges are:\n");
for(i=0;i<n;i++) {
Enter number of points: 4
0 0
0 4
4 4
4 0
Hull edges are:
=== Code Execution Successful ===
GoDaddy
Find Yours
Search
ENG
IN
13:39
01-08-2026